Madhav Jadav

Senior Full Stack Engineer with Blockchain Experience

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Professional Summary

STEM graduate (B.Tech-M.Tech Integrated Computer Science and Engineering, National Forensic Sciences University Delhi, 2021–2026) with 18 months project-based software engineering experience in backend development, REST APIs, and distributed systems for financial markets. Strong computer science fundamentals in data structures, algorithms, system design, and object-oriented and functional programming in Rust, Go, Python, Java, C++, C#, and JavaScript (Node.js). Built low-latency REST APIs serving 5K–10K requests per second with under 12ms latency, containerized microservices with automated testing and CI/CD, and a custom blockchain in Rust (SHA-256, Proof-of-Work) for digital asset custody logging. Experienced across full software development lifecycle: design, development, integration, automated testing, and maintenance in fast-moving release cycles. Collaborated with cross-functional stakeholders including quants, product, algo developers, infrastructure, and support. Genuine interest in crypto, digital assets, trading, execution and order management systems (OMS/EMS), portfolio and risk management. Experienced in backend development, blockchain technology, and high-performance microservices with a proven track record in securing systems and optimizing for speed and efficiency.

Technical Skills

Languages: Rust, Go (Golang), Python, Java, C++, C#, JavaScript/TypeScript (Node.js), SQL
Backend and APIs: REST APIs, Microservices, Event-Driven Architecture, System Design, gRPC, GraphQL, FastAPI, Actix-web, Node.js, JWT, RBAC, Single API Integration
Data and Infrastructure: PostgreSQL, MongoDB, Redis, Apache Kafka, Docker, Kubernetes, Helm, GitHub Actions (CI/CD), AWS EC2, MLflow
Engineering Practices: Software Development Lifecycle (SDLC), Design, Development, Integration, Automated Testing, Maintenance, Agile, Fast-Moving Release Cycle, Continuous Improvement, Stakeholder Collaboration, Code Review
FinTech and Digital Assets: Blockchain, SHA-256, Proof-of-Work, Digital Assets, Crypto, Trading, Execution, Order Management System (OMS/EMS), Portfolio Management, Risk Management, Market Connectivity, Secure SDLC, OWASP Top 10
Computer Science Fundamentals: Data Structures, Algorithms, Object-Oriented Programming, Functional Programming, Distributed Systems, Computer Networks, Operating Systems

Education

National Forensic Sciences University (NFSU), Delhi, India Nov 2021 – May 2026
B.Tech-M.Tech Integrated Computer Science and Engineering (Cyber Security) – STEM *CGPA: 7.5/10.0*

- Graduated 2026. Coursework: Data Structures, Algorithms, System Design, DBMS, Operating Systems, Computer Networks, Distributed Systems, Cryptography, OOP.
- Programming in Rust, Go, Python, Java, C++, JavaScript (Node.js). Strong analytical and problem-solving skills.

Premdhara Shishu Vihar High School, Gandhinagar, India – HSC (Class XII) 2021

- State Government, Gujarat. Physics, Chemistry, Maths.

Modern High School, Gandhinagar, India – SSC (Class X) 2019

Software Engineering Experience

Backend Software Engineer (Project-Based) – REST APIs, Microservices, Digital Assets Jan 2023 – Present
Self-Directed Engineering, Open Source and Hackathons | Delhi, India

- Develop and enhance backend software across full technology stack: design, development, integration, automated testing, and maintenance in fast-moving release cycles.
- Build and maintain features with business stakeholders: low-latency REST APIs serving 5K–10K requests per second with under 12ms latency using FastAPI, Redis caching, and PostgreSQL.
- Deliver trading-adjacent solutions: custom blockchain ledger in Rust (Actix-web, SHA-256, Proof-of-Work) modelling immutable execution logging, custody, and risk management workflows for digital assets.
- Work with cross-functional collaborators (product, algo/ML developers, infrastructure, support) to ship containerized microservices on Docker and Kubernetes with Helm, Kafka event streaming, and GitHub Actions CI/CD (deployment time cut from 2 hrs to 7 mins).
- Leverage new technologies with continuous improvement: gRPC, GraphQL, event-driven design, RBAC with Open

Policy Agent, MFA/JWT authentication with 99.99% uptime and zero unauthorized access in penetration tests.

Hackathon Engineer and CTF Competitor – Algo Development, Rapid Prototyping 2023 – 2025
Unstop, Devfolio, RedFox Security, SRM University | India

- Competed in 10+ hackathons and CTFs; built secure backend and AI/ML prototypes under deadlines, collaborating with product-minded stakeholders.

Projects

RhinoChain: Secure Biometric Blockchain System | Rust, Actix-web, SHA-256, Proof-of-Work, REST API Oct 2024 – Jan 2025
github.com/jadavmadhavkumar/rhino_chain

- Developed custom blockchain in Rust for immutable storage of hashed digital assets using SHA-256 and Proof-of-Work consensus – directly relevant to digital asset trading, execution logging, and risk management infrastructure.
- Built and documented single REST API for block addition, chain validation, and real-time updates; modular architecture with persistent storage and automated testing for cryptographic integrity.
- Applied system design, data structures, and algorithms for tamper-proof, privacy-preserving backend workflows and market connectivity concepts.

Secure Authentication Microservice | Node.js, JWT, MFA, RBAC, Kubernetes, Helm Sep 2023 – Dec 2023

- Engineered stateless backend microservice handling 10K RPS with JWT + TOTP MFA; 0% unauthorized access in penetration tests; 99.99% uptime with auto-scaling.
- Implemented RBAC with dynamic policies (Open Policy Agent) for 5+ roles; deployed via Docker, Kubernetes, Helm with full SDLC: design, integration, automated testing, maintenance.

ML-Powered Backend Engine (FastAPI + Redis) | Python, Scikit-learn, FastAPI, Redis, PostgreSQL Mar 2024 – Jun 2024

- Developed real-time backend classifier (99.2% accuracy on 1M records) exposed via FastAPI with Redis caching – under 12ms latency at 5K QPS; integrated into SSO portal, reducing weak inputs by 73%.
- Demonstrates low-latency API design, performance optimization, and continuous improvement applicable to execution and portfolio systems.

Real-Time Streaming Backend | Python, OpenCV, Kafka, Docker, GitHub Actions Dec 2022 – Mar 2023

- Architected event-driven streaming pipeline (30 FPS, 98.7% accuracy) with Apache Kafka and Docker; automated CI/CD reducing deployment from 2 hrs to 7 mins.

DcGAN Image Synthesis API | PyTorch, MLflow, AWS EC2 Oct 2023 – Jan 2024

- Trained DcGAN (generator loss 0.014, FID improved 42%) with MLflow tracking; deployed scalable REST API on AWS Spot Instances, cutting inference cost 68%.

Achievements

- **First Prize – Verge Technical Fest, SRM University (2025):** Top position for innovation, technical excellence, and problem-solving under deadlines. LinkedIn
- **Second Prize – CyberStorm CTF, RedFox Security (2025):** National cybersecurity competition: cryptography, reverse engineering, web exploitation. LinkedIn
- **Top Performer – 10+ National Hackathons, Unstop/Devfolio (2023–2025):** Built secure backend and AI/ML prototypes for rapid release cycles.

Certifications

- **Introduction to Generative AI** – Google Cloud Skills Boost (2023)
- **Cyber Apocalypse CTF** – Hack The Box, Global CTF (2023, 2024): web exploitation, cryptography, reverse engineering; secure SDLC practice.
- **Build Greeting App with React JS** – Coding Ninjas (2023)
- **Tata Data Visualization (TCS Forage)** – Executive dashboards and client-facing insights (Jan 2025)